

All-dimensional coexistency criteria for Hilbert-space effects

Candidate full solution packet for arXiv:2008.08559

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Status. Candidate full solution, likely valid, subject to expert review. Questions 6.2 and 6.3 are answered affirmatively for arbitrary Hilbert spaces. Question 6.4 receives an exact dimension-free support-functional characterization.

1 Source and the three questions

The source is G. P. Gehér and P. Šemrl, *Coexistency on Hilbert space effect algebras and a characterisation of its symmetry transformations*, arXiv:2008.08559v2, later published in *Communications in Mathematical Physics* **379** (2020), 1077–1112 [2]. The questions occur in Section 6, page 27 of the locally rendered source PDF.

the first case. If we set $i = 1$, then Lemma 3.2 and (4.1) imply $\mu_j \geq \lambda_j$ for all $j = 2, \dots, n-1$. But on the other hand, setting $j = n$ implies $\mu_i \leq \lambda_i$ for all $i = 2, \dots, n-1$. Therefore we conclude $B = A$. Similarly, assuming the second case implies $B = A^\perp$.

(ii) $\implies$ (i): Assume (i) does not hold, then there exists a positive number ε such that $\sigma(A) \subseteq [0, 1-\varepsilon]$ or $\sigma(A^\perp) \subseteq [0, 1-\varepsilon]$. Suppose the first possibility holds, then $\frac{1}{1-\varepsilon}A \notin \{A, A^\perp\}$ and $\left(\frac{1}{1-\varepsilon}A\right)^\sim \subseteq A^\sim$. The second case is very similar. $\square$

We only proved the above lemma and Corollary 2.11 in the finite dimensional case. The following two questions would be interesting to examine:

Question 6.2. Does the statement of Corollary 2.11 remain true for general infinite dimensional Hilbert spaces?

Question 6.3. Does the statement of Lemma 6.1 hold if $\dim H \geq \aleph_0$?

Finally, our first main theorem characterises completely when $A^\sim = B^\sim$ happens for two effects A and B . However, we gave only some partial results about when $A^\sim \subseteq B^\sim$ occurs, e.g. Lemma 2.12

Question 6.4. How can we characterise the relation $A^\sim \subseteq B^\sim$ for effects A, B ?

We believe that a complete answer to this latter question would represent a substantial step towards the better understanding of coexistence.

References

- [1] T. Ando, Problem of infimum in the positive cone, *Analytic and geometric inequalities and applications*, 1–12. *Math. Appl.* **478**. Kluwer Acad. Publ., Dordrecht, 1999.

Let H be a complex Hilbert space with $\dim H \geq 2$, and

$$\mathcal{E}(H) = \{A \in \mathcal{B}(H) : A = A^*, 0 \leq A \leq I\}.$$

Effects A, C are *coexistent*, written $A \sim C$, when there are effects M, N, G with $A = M + G$, $C = N + G$, and $M + N + G \leq I$. Put

$$A^\sim = \{C \in \mathcal{E}(H) : C \sim A\}, \quad A^\perp = I - A.$$

For a rank-one projection P , its strength in A is

$$\Lambda(A, P) = \max\{t \geq 0 : tP \leq A\},$$

and the source proves

$$tP \sim A \iff t \leq F_A(P) := \Lambda(A, P) + \Lambda(I - A, P). \quad (1)$$

The three source questions ask whether the finite-dimensional classification from equality of F_A extends to infinite dimensions, whether $0, 1 \in \text{spec}(A)$ characterizes minimal coexistence neighborhoods in infinite dimensions, and how arbitrary inclusion $A^\sim \subseteq B^\sim$ can be characterized.

2 Proof intuition

There are three independent mechanisms.

First, away from endpoint eigenspaces, strengths are reciprocals of inverse quadratic forms:

$$F_A(P_x) = \frac{1}{\|A^{-1/2}x\|^2} + \frac{1}{\|(I - A)^{-1/2}x\|^2} \quad (\|x\| = 1).$$

On any finite-dimensional real subspace of dimension at least three, these are reciprocals of positive-definite irreducible quadratic polynomials. Unique factorization recovers the unordered pair of inverse forms. A common form core patches the local choices globally, while endpoint eigenspaces are detected because their rank-one projections have strength sum one.

Second, inclusion $B^\sim \subseteq A^\sim$ implies that A and B commute. Localizing vectors near two points of their joint spectrum converts inclusion into a two-coordinate strength inequality. Points over spectral values 0 and 1 force opposite endpoint values for B ; comparison of either endpoint with an arbitrary joint spectral point forces both $b \geq a$ and $b \leq a$.

Third, $A^\sim$ is exactly the Minkowski sum $[0, A] + [0, I - A]$. Its support function against a finite-rank self-adjoint test operator has a closed positive-part trace formula. Hahn–Banach separation therefore turns arbitrary coexistence inclusion into an explicit family of scalar inequalities.

3 Two elementary lemmas

For an effect T , write

$$\mathcal{D}_T = \text{Ran}(T^{1/2}), \quad q_T(x) = \|T^{-1/2}x\|^2 \quad (x \in \mathcal{D}_T),$$

where the inverse is taken on $(\ker T)^\perp$. The Busch–Gudder strength formula gives

$$\Lambda(T, P_x) = \begin{cases} \|x\|^2 / q_T(x), & x \in \mathcal{D}_T, \ x \neq 0, \\ 0, & x \notin \mathcal{D}_T, \end{cases} \quad (2)$$

where P_x projects onto $\mathbb{C}x$ $[1, 2]$.

Lemma 1 (reciprocal quadratic rigidity). *Let V be a real vector space of finite dimension at least three, and let q_1, q_2, r_1, r_2 be positive-definite quadratic forms on V . If q_1, q_2 are not proportional and*

$$\frac{1}{q_1(x)} + \frac{1}{q_2(x)} = \frac{1}{r_1(x)} + \frac{1}{r_2(x)} \quad (x \neq 0),$$

then $\{r_1, r_2\} = \{q_1, q_2\}$.

It is impossible on a nonempty open subset of V to have

$$\frac{1}{q_1} + \frac{1}{q_2} = \frac{1}{r}$$

with r a positive-definite quadratic form.

Proof. A positive-definite quadratic form in at least three real variables is irreducible over $\mathbb{R}$. Clearing denominators in the first identity gives

$$(q_1 + q_2)r_1r_2 = (r_1 + r_2)q_1q_2.$$

The irreducible forms q_1, q_2 are coprime, and neither divides $q_1 + q_2$. Unique factorization therefore matches each q_i , up to a positive scalar, with one r_j . Substitution in the reciprocal identity makes both scalars one. The single-reciprocal identity would give $r(q_1 + q_2) = q_1q_2$; the same divisibility argument forces r proportional to one q_i and then forces q_1, q_2 proportional, a contradiction. An identity on an open set is a polynomial identity after denominators are cleared. $\square$

Lemma 2 (one-variable pole rigidity). *Let $0 < c < 1$ and $p, q, r > -1$. If*

$$\frac{c}{1+sp} + \frac{1-c}{1+sq} = \frac{1}{1+sr} \quad (0 < s < 1),$$

then $p = q = r$.

Proof. After clearing denominators, comparison of the coefficients of s and s^2 says that the unordered pair $\{p, q\}$ equals

$$\{r, cq + (1-c)p\}.$$

If $p \neq q$, the second displayed number lies strictly between them and cannot be either member of the pair. Hence $p = q$, and then $r = p$. $\square$

We also use the standard variational identity

$$\langle Ty, y \rangle = \sup_{x \in \mathcal{D}_T} (2 \operatorname{Re} \langle y, x \rangle - q_T(x)) \quad (T \geq 0). \quad (3)$$

It follows directly by putting $x = T^{1/2}z$ and completing the square; the formula remains valid when T has a kernel.

4 Question 6.2: rank-one coexistency determines the effect

Theorem 3. *For arbitrary $A, B \in \mathcal{E}(H)$, the following are equivalent:*

1. $F_A(P) = F_B(P)$ for every rank-one projection P ;
2. $A^\sim \cap \{tP : 0 < t \leq 1, P \in \mathcal{P}_1(H)\} = B^\sim \cap \{tP : 0 < t \leq 1, P \in \mathcal{P}_1(H)\}$;
3. either A, B are scalar effects, or $B = A$, or $B = I - A$.

Thus the answer to Question 6.2 is affirmative, with no separability assumption.

Proof. The equivalence of (1) and (2) follows from (1); (3) implies both immediately. Assume (1).

The endpoint-free case. Suppose first that $\ker A = \ker(I - A) = \{0\}$. Set

$$\mathcal{D} = \mathcal{D}_A \cap \mathcal{D}_{I-A}.$$

This is a dense complex-linear subspace, since it contains the ranges of all spectral cuts $E_A([1/n, 1 - 1/n])$. If A is non-scalar, the forms q_A and q_{I-A} are not proportional on $\mathcal{D}$: proportionality, followed by closure on the spectral-cut core, would give $A^{-1} = c(I - A)^{-1}$ and hence $A = (1 + c)^{-1}I$.

If A is scalar, then $F_A(P) = 1$ for every P . Thus every rank-one projection coexists with B , and therefore commutes with B ; hence B is scalar. We may consequently assume for the remainder of this case that A is non-scalar.

Choose a finite-dimensional real $V \subset \mathcal{D}$, $\dim_{\mathbb{R}} V \geq 3$, on which the two forms are not proportional. For every nonzero $x \in V$, the left side of

$$\frac{1}{q_A(x)} + \frac{1}{q_{I-A}(x)} = \Lambda(B, P_x)/\|x\|^2 + \Lambda(I - B, P_x)/\|x\|^2 \quad (4)$$

is positive. Hence

$$V \subseteq \mathcal{D}_B \cup \mathcal{D}_{I-B}.$$

The intersections of the two domains with V are real linear subspaces, and a vector space over an infinite field is not the union of two proper subspaces. Thus one domain contains V . If the other did not, then on the nonempty open complement of its intersection with V , equation (4) would express the sum of two reciprocal nonproportional quadratic forms as one reciprocal quadratic form, contradicting Lemma 1. Consequently V lies in both domains, and Lemma 1 gives

$$\{q_B|_V, q_{I-B}|_V\} = \{q_A|_V, q_{I-A}|_V\}. \quad (5)$$

Fix one such base space V_0 . For any $x \in \mathcal{D}$, apply the same argument to the real span of V_0 and x . Because the two forms remain nonproportional on the enlarged space, (5) holds there. Its orientation must agree with the one on V_0 . We therefore have, on all of $\mathcal{D}$, either

$$q_B = q_A, \quad q_{I-B} = q_{I-A}, \quad (6)$$

or the two equalities with A and $I - A$ exchanged.

The spectral cuts $E_A([1/n, 1 - 1/n])$ show that $\mathcal{D}$ is a form core for both inverse forms of A . In the direct orientation, (3) and (6) yield

$$\langle By, y \rangle \geq \langle Ay, y \rangle, \quad \langle (I - B)y, y \rangle \geq \langle (I - A)y, y \rangle$$

for every $y \in H$: the supremum for B is over a domain containing $\mathcal{D}$, while the supremum over the core $\mathcal{D}$ equals the value for A . Hence $B \geq A$ and $B \leq A$, so $B = A$. The exchanged orientation similarly gives $B = I - A$.

Endpoint eigenspaces. Let

$$H_0 = \ker A, \quad H_1 = \ker(I - A), \quad H_g = (H_0 \oplus H_1)^\perp,$$

and write $A_g = A|_{H_g}$. For every unit $x \in H_0 \cup H_1$, $F_A(P_x) = 1$. Therefore P_x coexists with B and, since P_x is a projection, it commutes with B [2, Lemma 2.1(b)]. It follows that B reduces H_0, H_1, H_g and is scalar on each nonzero endpoint space:

$$B = c_0 I_{H_0} \oplus B_g \oplus c_1 I_{H_1}.$$

On H_g , the already proved endpoint-free case says $B_g = A_g$ or $I - A_g$ whenever A_g is non-scalar. If $A_g = eI$ is scalar, equality of the restricted strength sums makes every rank-one projection on H_g commute with B_g , so $B_g = dI$ is scalar.

Assume first $H_0 \neq \{0\}$ and $H_g \neq \{0\}$, and choose unit vectors $x \in H_0$ and

$$y \in \mathcal{D}_{A_g} \cap \mathcal{D}_{I-A_g}.$$

For $z_s = \sqrt{1-s}x + \sqrt{s}y$, $0 < s < 1$, formula (2) gives

$$F_A(P_{z_s}) = \frac{1}{1 + s(q_{I-A_g}(y) - 1)}. \quad (7)$$

If A_g is non-scalar and $B_g = A_g$, then $0 < c_0 < 1$ would turn the left side for B into

$$\frac{c_0}{1 + s(c_0 q_{A_g}(y) - 1)} + \frac{1 - c_0}{1 + s((1 - c_0)q_{I-A_g}(y) - 1)}.$$

Lemma 2, compared with (7), would imply $(1 - c_0)q_{I-A_g}(y) = q_{I-A_g}(y)$, impossible. Thus $c_0 \in \{0, 1\}$. If $c_0 = 1$, equation (7) forces $q_{A_g} = q_{I-A_g}$ on a common core, hence $A_g = I/2$, again impossible. Therefore $c_0 = 0$. The alternative $B_g = I - A_g$ gives symmetrically $c_0 = 1$.

If $A_g = eI$ and $B_g = dI$, restriction to $\mathbb{C}x \oplus \mathbb{C}y$ is the two-dimensional diagonal situation already proved in Corollary 2.11 of the source: equality of all rank-one strength sums gives $(c_0, d) = (0, e)$ or $(1, 1 - e)$. Thus the same global alternatives hold.

If both H_0 and H_1 are nonzero, their two-dimensional compression forces $(c_0, c_1) = (0, 1)$ or $(1, 0)$, consistently with the orientation on H_g . If $H_g = 0$, this argument is exactly the projection case. If $H_0 = 0$ but $H_1 \neq 0$, apply the preceding argument to $I - A$, noting that $F_{I-A} = F_A$. The only remaining cases are scalar. This proves (3). $\square$

5 Question 6.3: minimal coexistence neighborhoods

Lemma 4 (joint spectral transfer). *Let A, B be commuting effects and suppose $F_B(P) \leq F_A(P)$ for every rank-one projection P . Let X be the Gelfand spectrum of $C^*(I, A, B)$ and write $a, b \in C(X)$ for the Gelfand transforms. Then for all $s, t \in X$ and $0 < \theta < \pi/2$,*

$$T_{b(s), b(t)}(\theta) \leq T_{a(s), a(t)}(\theta), \quad (8)$$

where

$$T_{u,v}(\theta) = \left(\frac{\cos^2 \theta}{u} + \frac{\sin^2 \theta}{v} \right)^{-1} + \left(\frac{\cos^2 \theta}{1-u} + \frac{\sin^2 \theta}{1-v} \right)^{-1},$$

with a reciprocal equal to zero when its denominator is infinite.

Proof. For distinct s, t , choose shrinking disjoint joint spectral neighborhoods on which both a and b differ from their values at the centers by at most ε . Their joint spectral projections in $C^*(A, B)''$ are nonzero: otherwise a nonzero continuous function supported in such a neighborhood would be represented by zero, contradicting faithfulness. Choose unit vectors in the two orthogonal reducing ranges and form their θ -mixture. On each range, A and B are squeezed in operator order between the corresponding scalar values plus or minus ε . Apply (2) separately to the effect and its complement. Letting $\varepsilon \downarrow 0$ gives (8). The case $s = t$ follows by continuity or is immediate. $\square$

Theorem 5. *For every effect A on an arbitrary Hilbert space H , the following are equivalent:*

1. $0, 1 \in \text{spec}(A)$;
2. there is no $B \in \mathcal{E}(H)$ such that $B^\sim \subsetneq A^\sim$.

Thus Question 6.3 has an affirmative answer, without separability.

Proof. Assume $0, 1 \in \text{spec}(A)$ and $B^\sim \subseteq A^\sim$. Lemma 2.12(a) of the source, with the roles there chosen as B, A , gives $A \in B''$; in particular A and B commute [2]. By (1), inclusion gives $F_B \leq F_A$.

Use the notation of Lemma 4. Since $a(X) = \text{spec}(A)$, choose $s_0, s_1 \in X$ with $a(s_0) = 0$ and $a(s_1) = 1$. For every interior θ ,

$$T_{0,1}(\theta) = 0.$$

Equation (8) and nonnegativity imply

$$\{b(s_0), b(s_1)\} = \{0, 1\}.$$

Suppose first $b(s_0) = 0$ and $b(s_1) = 1$. For arbitrary $s \in X$, comparison with s_0 gives

$$T_{0,b(s)}(\theta) \leq T_{0,a(s)}(\theta).$$

For interior θ ,

$$T_{0,u}(\theta) = \left(\cos^2 \theta + \frac{\sin^2 \theta}{1-u} \right)^{-1},$$

which is decreasing in u ; hence $b(s) \geq a(s)$. Comparison with s_1 gives

$$T_{1,b(s)}(\theta) \leq T_{1,a(s)}(\theta), \quad T_{1,u}(\theta) = \left(\cos^2 \theta + \frac{\sin^2 \theta}{u} \right)^{-1},$$

and this is increasing in u , so $b(s) \leq a(s)$. Thus $b = a$ on X and $B = A$. The reversed endpoint orientation gives $b = 1 - a$ and $B = I - A$. In either case the inclusion is equality.

Conversely, if an endpoint is missing and A is non-scalar, the scaling argument from the proof of Lemma 6.1 in the source applies verbatim in every dimension. For example, if $\text{spec}(A) \subset [0, 1 - \varepsilon]$, then a suitable nontrivial rescaling $C = (1 - \varepsilon)^{-1}A$ satisfies $C^\sim \subsetneq A^\sim$ by convexity and Theorem 2.9 of the source. The case with the other endpoint is obtained from $I - A$. If A is scalar, $A^\sim = \mathcal{E}(H)$ and any nontrivial projection has a proper coexistence neighborhood. This proves the contrapositive. $\square$

6 Question 6.4: an exact support-functional characterization

For a finite-rank self-adjoint operator T , define

$$h_A(T) = \text{Tr}((A^{1/2}TA^{1/2})_+) + \text{Tr}(((I - A)^{1/2}T(I - A)^{1/2})_+), \quad (9)$$

where $S_+ = (|S| + S)/2$.

Theorem 6. *For arbitrary effects A, B on any Hilbert space,*

$$A^\sim \subseteq B^\sim \iff h_A(T) \leq h_B(T) \text{ for every finite-rank self-adjoint } T. \quad (10)$$

Equivalently, inclusion is determined by explicit positive-part trace inequalities on all finite-dimensional compressions.

Proof. The coexistence decomposition used in the source is precisely

$$A^\sim = [0, A] + [0, I - A], \quad (11)$$

where $[0, A] = \{M : 0 \leq M \leq A\}$. Indeed, $C = M + N$ with $M \leq A$ and $N \leq I - A$ is equivalent to $C \sim A$.

Fix finite-rank self-adjoint T . By Douglas factorization, every $M \in [0, A]$ has the form $M = A^{1/2}KA^{1/2}$ with $0 \leq K \leq I$ on the support of A , and conversely every such operator belongs to $[0, A]$. Therefore

$$\begin{aligned} \sup_{M \in [0, A]} \text{Tr}(TM) &= \sup_{0 \leq K \leq I} \text{Tr}(KA^{1/2}TA^{1/2}) \\ &= \text{Tr}((A^{1/2}TA^{1/2})_+), \end{aligned}$$

the maximum being attained at the positive spectral projection. Support functions add under Minkowski sums, so (11) gives

$$\sup_{C \in A^\sim} \text{Tr}(TC) = h_A(T).$$

Inclusion implies the inequalities in (10). Conversely, $B^\sim$ is convex and WOT compact [2, Lemma 2.6]. If some $C \in A^\sim$ were outside $B^\sim$, real WOT Hahn–Banach separation would provide a WOT-continuous real functional strictly separating C from $B^\sim$. Every such functional on the self-adjoint part of $\mathcal{B}(H)$ is $X \mapsto \text{Tr}(TX)$ for a finite-rank self-adjoint T . This would give $h_A(T) > h_B(T)$, contradicting the assumed inequalities. Hence inclusion holds. $\square$

The criterion (10) is a complete answer in convex-dual form. It is not claimed to be a simpler pointwise functional-calculus classification of all pairs (A, B) ; rather, it replaces quantified coexistence decompositions by explicit scalar spectral expressions and is valid without dimension or separability restrictions.

7 Verification and novelty status

Internal checks. The proof was checked against the following edge cases: scalar effects; non-trivial projections; one-dimensional generic spectral summands; effects with continuous spectrum touching both endpoints; and nonseparable spaces. The file `code/check_identities.py` symbolically verifies the pole identity used in Lemma 2 and numerically checks the support-function maximizer and the direct/complement strength identities on random finite matrices. These checks are diagnostics, not substitutes for the proof.

Novelty search. The bounded search covered the run indexes, the locally parsed arXiv corpus, the exact title and arXiv id, the source authors, and phrases combining “coexistence/coexistence”, “rank-one effects”, “infinite dimensional”, and “minimal coexistence neighborhood”. Later local papers citing Gehér–Šemrl were inspected at the citation context; none stated an answer to Questions 6.2–6.4. A fresh external-network search was unavailable in the current environment. Accordingly the proof verdict is *likely valid*, while novelty confidence is *moderate* pending a human literature refresh.

Recommended expert review. Check especially (i) the form-domain union and orientation-patching argument in Theorem 3, (ii) the passage from joint spectral neighborhoods to (8), and (iii) the WOT separation claim in Theorem 6. If these pass, the packet resolves both infinite-dimensional questions and supplies an exact answer to the general inclusion question.

References

- [1] P. Busch and S. Gudder, Effects as functions on projective Hilbert space, *Letters in Mathematical Physics* **47** (1999), 329–337.
- [2] G. P. Gehér and P. Šemrl, Coexistence on Hilbert space effect algebras and a characterisation of its symmetry transformations, arXiv:2008.08559v2; *Communications in Mathematical Physics* **379** (2020), 1077–1112.